

Lodestar

Unity SDK Documentation

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1. Introduction

The **Lodestar Unity SDK** provides a seamless way to integrate 6DoF spatial tracking data from Lodestar hardware into Unity applications. Designed for XR, simulation, robotics, and virtual production workflows, the SDK leverages Unity's Input System to provide high-performance, low-latency tracking with minimal setup.

2. Requirements

- **Unity Version:** 2022.3 LTS or later
- **Unity Input System Package:** v1.7.0 or later
- **Hardware:** [Lodestar Hub \(and paired Pucks\)](#) connected via USB

3. Installation

1. Create or open a Unity project.
2. Ensure the Unity Input System is installed via Package Manager:
 - Go to `Window > Package Manager`
 - Search for **Input System**, install version 1.7.0 or newer.
 - Unity will prompt to restart — accept.
3. Import the Lodestar Unity SDK `.unitypackage` file into the project.
4. After import, the SDK is ready for use.

4. Setup Guide

Option A: Use the Sample Scene

1. Navigate to `Lodestar SDK/Scenes/`
2. Open `LodestarSample.unity`
3. Connect the Lodestar Hub via USB
4. Enter Play mode - tracker GameObjects will appear in the scene.

Option B: Integrate into Your Own Scene

1. In your scene, drag the prefab `LodestarManager` from `Lodestar/Prefabs/` into the hierarchy.
2. (Optional) Assign:
 - **Tracker Object:** Your own prefab to represent trackers (must have `LodestarTracker.cs` component).
 - **Tracker Origin:** A Transform to define the origin of the tracking space.
3. Enter Play mode with the Lodestar Hub connected.

The system will auto-detect all connected trackers and spawn a GameObject for each one, parented to the origin.

5. Using the SDK

Once the system is running:

- Each connected tracker is instantiated as a GameObject with the `LodestarTracker.cs` script.
- These tracker objects will automatically update their transform based on real-time position and rotation data.
- The default prefab includes a simple visual representation, but you can customize this.

No code is required for basic tracking.

6. Script Reference

`LodestarManager.cs`

Purpose:

Initializes the SDK, handles detection and spawning of tracker objects at runtime.

Key Fields:

```
GameObject trackerObject
```

Prefab to instantiate per tracker (must contain `LodestarTracker.cs`).

```
Transform trackerOrigin
```

The parent transform under which trackers are spawned. Defaults to manager's transform.

```
Dictionary<int, GameObject> trackerRegistry
```

Dictionary of all tracker GameObjects in the scene by ID. Note this ID is assigned by the Unity Input System and is different from the Puck ID.

LodestarTracker.cs

Purpose:

Holds all device data and updates transform in real time.

Public Fields / Properties:

```
int id;  
int battery;  
Status status;  
bool tracking;  
Pose pose;
```

Accessing Trackers via Code:

```
LodestarTracker tracker = FindObjectOfType<LodestarTracker>();  
Debug.Log(tracker.id);
```

You can search or cache trackers via `FindObjectsOfType<LodestarTracker>()` or by accessing the manager's `trackerRegistry` Dictionary.

7. Customization

- **Custom Tracker Prefabs:**

Replace the visual model or add additional components to your prefab.

Important: Must have the `LodestarTracker` component attached.

- **Changing the Tracking Origin:**

Assign the `trackerOrigin` field in `LodestarManager` to any transform.

All tracker objects will be positioned relative to this transform.

8. Troubleshooting

Q: No tracker objects appear in Play mode?

- Make sure the Lodestar Hub is connected via USB.
- Ensure Unity Input System is enabled.
- Verify `LodestarManager` is present in the scene.

Q: My custom tracker prefab isn't working.

- Confirm it contains the `LodestarTracker` script.
- Check for missing components or conflicting scripts.

9. Contact & Support

Dashboard: lodestar.tools

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